(a) Problem

Heterogeneous LLM Annotation Error

LLM Classifier

Subgroup A
$\pi = 0.05$

5×

Subgroup B
$\pi = 0.25$

Error rates vary up to 5× across subgroups

(b) Theory

Why Global Correction Fails

Pooled CM ($\bar{C}$)
One matrix for all

Subgroup A
Over-corrected ↗

Subgroup B
Under-corrected ↘

$$bias(z, s) = \frac{\delta_s \cdot \Delta\tau}{1 - 2\bar{\pi}}$$

⚠ **Sign Reversal: 14.9% of configs**

(c) Solution: EC-HTE

Per-Subgroup Correction Pipeline

Step 1: Hierarchical Bayes
Per-subgroup $\hat{C}_s$ via partial pooling (α_0)

Step 2: Stratified Inversion
$\hat{C}_s^{-1}$ with κ-guard ($\kappa > 100 \rightarrow$ fallback)

Step 3: Corrected CATE
$\hat{\tau}(x) \pm SE_{corr}$ via CausalForest / DIM

Eliminates sign reversal; composable with PPCI

(d) Key Results

Validated across 8 LLMs, 4 NLP tasks

7.4×
|bias| reduction

14.9% → 0%
Sign reversal eliminated

≥ 93%
CI coverage maintained

+ Δτ-free diagnostic (Prop. 5)